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Interfacial Dipoles Boost Open-Circuit Voltage of Tin Halide Perovskite Solar Cells

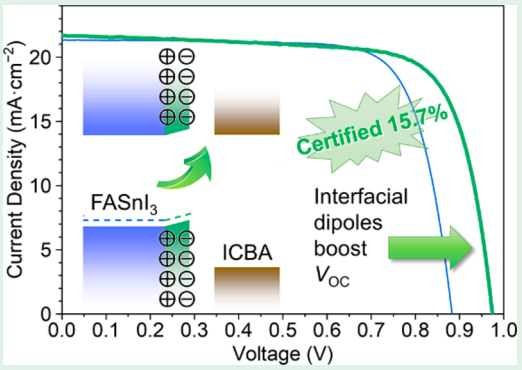
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ABSTRACT: **Tin halide perovskites are promising materials for real-world photovoltaic applications, and there is large room for their performance to grow. We improved energy-level alignment between the perovskite and electron transport layers by introducing 4-fluorophenethylamine hydrobromide as an interfacial dipole. Thus, the open-circuit voltage of $FASnI_3$ solar cells increased to 0.974 V and their power conversion efficiency reached 15.7%, representing a significant advance in this field.**



There has been remarkable progress in the photovoltaic performance of perovskite solar cells, with energy conversion efficiencies exceeding 26%.[1,2] While top device parameters are accomplished by lead-based materials, tin halide perovskites (THPs) have also garnered interest due to their stable crystal structure, suitable bandgap, high charge-carrier mobility, and environmentally friendly characteristics.[3−6] However, THPs feature distinct energetics and crystallization kinetics from their lead counterparts and require specially adapted formulations and components for device fabrication. For example, introducing 2D structures[7] and Lewis-base ligands[8] significantly enhances the quality of $FASnI_3$ (FA^+ = formamidinium) films. Bisadducts of fullerene C_{60} are found to be excellent electron transport materials for high open-circuit voltages (V_{oc}).[9,10] In the past three years, the power conversion efficiency (PCE) of THP solar cells has arrived at a plateau of 14−15% (Figure 1a), only about half of the possible upper limit. Further performance enhancement will make THP solar cells particularly suitable for powering building-integrated, wearable, and healthcare applications.

Our earlier work[8] adopted the inverted solar cell structure ITO/PEDOT/perovskite/ICBA/Ag, where the PEDOT hole transport layer, perovskite active layer, ICBA electron transport layer, and Ag electrodes were deposited on ITO glass substrates in order. Incorporating organic ammonium salts as a 2D scaffold and substituted thioureas as ligands afforded smooth and compact $FASnI_3$ films and solar cells of 14.3% PCE. Herein, by using 4-fluorophenethylamine hydrobromide (FPEABr) as a molecular dipole, we better matched the energy levels of the perovskite and ICBA layers and unilaterally increased the device V_{oc} by a tenth, thus achieving a new PCE record of 15.7% (Figure 1b).

The molecular structure of FPEABr bears a dipole moment pointing from ammonium to the electronegative F at the other end (Figure 2a). Including 1.5 mol% of FPEABr in the perovskite layer had a negligible effect on its bandgap energy, E_g = 1.42 eV (Figure S1). Surprisingly, such a small amount boosted the device V_{oc} from 0.883 to 0.974 V. Considering that F atom(s) attached to the phenyl ring can exhibit notable surface effects,[11−13] we performed ultraviolet photoelectron spectroscopy (UPS) to determine the valence band position of the perovskite layer. In Figure 2b, the spectral data are plotted relative to the vacuum level and aligned to a cutoff energy of −21.22 eV, which is the deepest level that can be excited by He I photons. Linear extrapolation of the band edges gives valence band maximum (VBM) positions of −5.07 and −5.17 eV for $FASnI_3$ films with or without additional FPEABr, respectively. Figure S2 emphasizes an energy difference of 0.10 eV between the two cases.

Figure 2c depicts the device structure of solar cells, highlighting the role of FPEABr with its F atom exposed and the ammonium group anchored to the top surface of $FASnI_3$.

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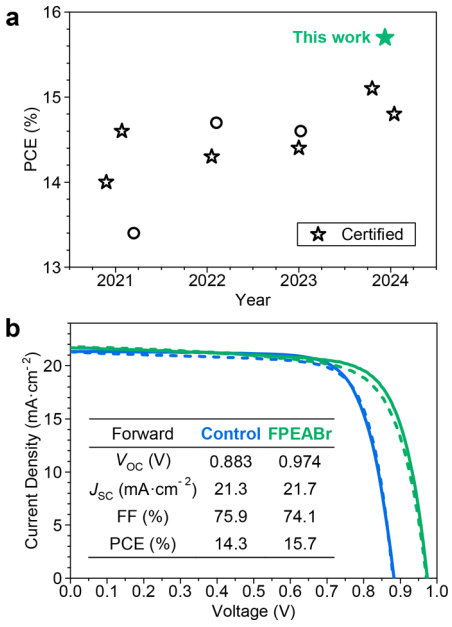


Figure 1. (a) Progress in the power conversion efficiency (PCE) of the best-performing tin halide perovskite solar cells, sorted by date of certification or publication. (b) Current density−voltage curves and performance parameters of devices with or without FPEABr.

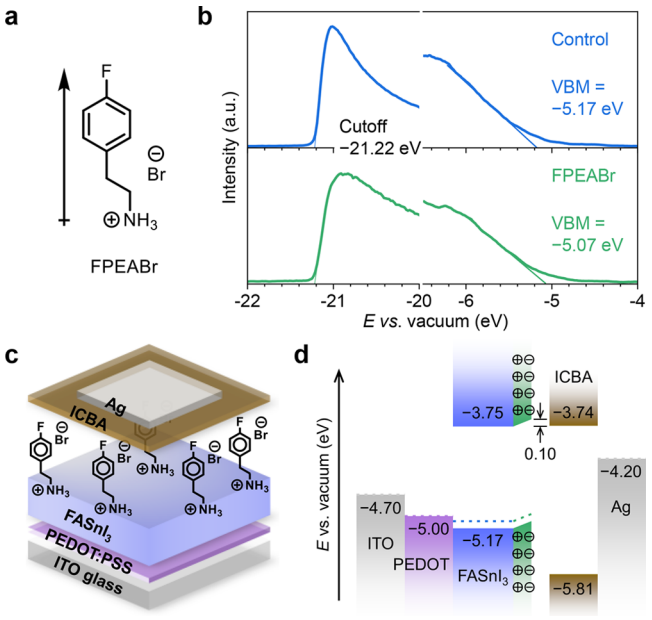


Figure 2. Interfacial dipoles due to extra FPEABr on the perovskite surface. (a) Molecular dipole of FPEABr. (b) Ultraviolet photoelectron spectra (UPS) of $FASnI_3$ films, indicating elevated valence band maximum (VBM) due to FPEABr. (c) Structure and (d) energy levels of the inverted solar cell structure. The interfacial dipoles (green) are assumed to lie between the perovskite and electron transport layers and lift the band edge energies by 0.10 eV.

From the known energy levels of charge transport materials,[9,14] as well as the measured VBM and E_g values of our $FASnI_3$ films, an energy band diagram is constructed in Figure 2d. In control devices, the conduction band minimum (CBM) of the perovskite layer is located at −3.75 eV, on par with the lowest unoccupied molecular orbital (LUMO) energy of ICBA. In the presence of FPEABr, the interfacial dipoles increase both the VBM and CBM of the perovskite layer by 0.10 eV, making the perovskite−ICBA junction energetically favorable for electron transport.[15]

Based on these preliminary characterization results, we propose a mechanism for the effect of FPEABr on V_{oc} enhancement. Because organic cations, Sn^{2+}, and I^- have balanced quantities in the precursor solution, a small excess of the weakly coordinating $FPEA^+$ and Br^- is expected to participate in $FASnI_3$ crystallization only at its later stage, eventually becoming surface-dominant. Hence, surfaces of the $FASnI_3$ layer were better protected to display a longer charge-carrier lifetime (Figure S3). However, the influence of FPEABr on the morphology of the perovskite layer was marginal (Figure S4). FPEABr dipoles at the perovskite−ICBA interface then facilitate the extraction of photoexcited electrons, which limit photovoltage generation as minority carriers in p-type $FASnI_3$. As a side effect, the interfacial dipoles led to a hysteresis phenomenon, and the reverse $J-V$ scan recorded a slightly reduced PCE of 15.1% (Figure 1b).

In conclusion, we successfully adjusted the energy band positions of the $FASnI_3$ perovskite layer using FPEABr and attained high device performances for lead-free perovskite solar cells. We propose that FPEABr exists primarily between the perovskite and electron transport layers, forming interfacial dipoles. Considering E_g = 1.42 eV for $FASnI_3$, V_{oc} = 0.974 V realized in this work is satisfactory but J_{sc} remains relatively low. Further studies on enlarging J_{sc} and diminishing hysteresis, as well as tracking device stability, are underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.4c00529.

Materials and methods, UPS and EQE data, PL decay kinetics, SEM images, and additional references (PDF)

Performance measurement report (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

2D, two-dimensional; CBM, conduction band minimum; EQE, external quantum efficiency; FF, fill factor; FPEABr, 4-fluorophenethylamine hydrobromide; ICBA, indium−C_{60} bisadduct; ITO, indium tin oxide; LUMO, lowest unoccupied molecular orbital; PCE, power conversion efficiency; PEDOT, poly(3,4-ethylenedioxythiophene); SEM, scanning electron microscopy; THP, tin halide perovskite; UPS, ultraviolet photoelectron spectroscopy; VBM, valence band maximum

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